

# Muhammad Bayoumi

Technical Office Team Leader at Saudi Pan Kingdom Company (SAPAC)

Riyadh, Saudi Arabia   [muhammad.bayoumi.ali@gmail.com](mailto:muhammad.bayoumi.ali@gmail.com)   [linkedin.com/in/muhammad-bayoumi-ali](https://www.linkedin.com/in/muhammad-bayoumi-ali)  
[muhammadbayoumi.github.io](https://muhammadbayoumi.github.io)

## Summary

Civil engineer bridging construction and data

Civil engineer with 7+ years of experience in tendering, technical office work, quantity surveying, and cost estimation on large-scale construction projects across Egypt and Saudi Arabia.

The mind behind mbiX — an Excel VSTO add-in designed to help teams pull and update data inside Microsoft Excel seamlessly — eliminating the friction of manual updates.

A self-motivated, ethics-driven engineer committed to continuous learning and to delivering real value to colleagues in ways that strengthen team spirit and collaboration. My ambition is to be part of leading positive change in the world.

## Experience

Technical office, tendering, and site roles since 2019

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| SAUDI PAN KINGDOM COMPANY (SAPAC)<br>Full-time · Riyadh, Saudi Arabia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                |
| Sep 2026 – Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Technical Office Team Leader<br>On-site                        |
| Feb 2025 – Aug 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Senior Technical Office Engineer                               |
| SAMCO-NATIONAL CONSTRUCTION COMPANY<br>Cairo, Egypt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                |
| Jan 2024 – Jan 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Senior Tendering, QS and Cost Estimation Engineer<br>Full-time |
| <div><div><ul style="list-style-type: none"><li>Study the tender document</li><li>Prepare the letter of bid or no bid</li><li>Review bid bond requests for tenders</li><li>Review and handle the ISO documents</li><li>Prepare the time schedule</li><li>Review the administrative submittal</li></ul></div><div><ul style="list-style-type: none"><li>Prepare site visit reports</li><li>Answer pre-qualification file requests</li><li>Read and understand customers' RFQs and specifications</li><li>Undertake a bid / no-bid exercise to decide whether to make a submission</li></ul></div></div> |                                                                |

- Understand the RFT, deciding timelines and responsibilities
- Manage tender activities on time
- Review drafts and make improvements
- Obtain feedback from the commercial team when tenders are unsuccessful, to improve future bids
- Coordinate stakeholders across departments to build a coherent offer and respect deadlines
- Tackle challenges met by stakeholders and help find solutions
- Assess the risks of each RFQ (client, planning, over cost, political), mitigate them, and comply with group internal procedures
- Write and prepare offers in compliance with the client's formal requirements
- Build profit and loss scenarios (worst and best case)

Jun 2021 – Dec 2023

### Technical Office Engineer

Full-time

- Study the project contract documents (specifications, drawings, BOQ, method of measurement, and contracts) and prepare requests for information (RFI) for anything missing, unclear, or conflicting
- Prepare shop drawings for overpass and underpass structures and submit them for the client consultant's approval before execution
- Prepare quantity surveying for projects and issue client invoices, including their supporting documents
- Issue submittals for all materials used in the projects
- Review subcontractors' shop drawings and coordinate their approval with the client consultant
- Monitor project budgets against the tendering budget with the cost control department, follow cash-in and cash-out, and compare labour and equipment productivity against budget to improve financial performance
- Conduct studies on variation orders during implementation
- Prepare cost breakdowns and the documents needed for price discussion committees
- SAP ERP user

Feb 2021 – Jun 2021

### Site Engineer

Internship

- Supervise all structural concrete construction activities of the Alferdos axis bridge
- Ensure execution according to approved drawings, design, and specifications

MBI

Self-employed

Nov 2020 – Jan 2021

### Contractor

Cairo and Qalyubia · Hybrid

- Prepare executive drawings, tender, and abstracts, and build an RC foundation carrying water pumps for HelioPower at Mostorod Water Station
- Build a footpath for ELSERAFY ENGINEERING LLC in Qalyub Central

May 2020 – Oct 2020

**Technical Office Engineer**

Banha, Al Qalyubiyah, Egypt

- Negotiate with clients on preparing construction reconciliation files — more than 100 clients served
- Survey facilities and draw as-built architectural and construction drawings
- Carry out structural design to ensure structural safety, and prepare the calculation notes
- Obtain approval of drawings and reports from consulting engineers, and follow up the safety certificate from the Engineers Association

**EGYPTIAN ARMED FORCES ENGINEERING AUTHORITY**

Full-time · Port Said, Egypt

Mar 2019 – Feb 2020

**Supervision & Technical Office Engineer**

East Port Said Naval Base

- Supervise infrastructure, construction, landscape, and finishing works to approved drawings, design, and specifications
- Supervise soil improvement by the CMC (Controlled Modulus Column) method
- Follow up performance rates and submit daily reports

**TRAINING**

Mar 2018 – May 2018

**Trainee · Bauer Foundation Corp.**

500500 Hospital, 57357 Hospital, and the Extensions of the Bar Association of Cairo

- Follow up construction of the diaphragm wall and its soil-nail tiebacks at 500500 Hospital
- Follow up construction of the diaphragm wall, its temporary anchors, and dewatering at 57357 Hospital
- Follow up jet-grouting soil improvement and dewatering at the Extensions of the Bar Association of Cairo

Jun 2015

**Trainee · The Arab Contractors**

- Practical workshops in construction: formwork, reinforcement, masonry, plaster, electrical, plumbing, woodwork, metalwork, and painting

## Projects

Roads, bridges, tunnels, and marine works, and the tools built along the way

**SAUDI PAN KINGDOM COMPANY (SAPAC)**

Feb 2025 – Present

**Second Southern Ring Road (SSRR)**

Contract period 5 Aug 2024 – 4 Aug 2028

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| SAMCO-NATIONAL CONSTRUCTION COMPANY                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Jul 2023 – Apr 2024</p> <p>Bridge at the intersection of the 3rd of July Corridor and Hasaballah Al-Kafrawy Corridor</p> | <p>May 2023 – Oct 2023</p> <p>Tunnel at the intersection of the Sphinx Corridor and the Dabaa Corridor</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>Dec 2022 – May 2023</p> <p>Tunnel No. 1 within the coastal road</p>                                                      | <p>May 2022 – Sep 2022</p> <p>Egyptian Rural Development Project (Decent Life), Phase One — Fayoum Governorate, Atsa Center</p> <p>Replacing and renewing irrigation and drainage bridges — 30 bridges in four regions</p> <ul style="list-style-type: none"> <li>• First region: Qalhana – Shadmouh, 4 bridges (May – Aug 2022)</li> <li>• Second region: Tawil – Minyat Al-Hait – Abu Jendir – Abu Sir, 19 bridges (Jun – Sep 2022)</li> <li>• Third region: Qasr al-Basil – Tatoun, 4 bridges (Jun – Sep 2022)</li> <li>• Fourth region: Al-Gharq – Al-Hajar, 3 bridges (May – Aug 2022)</li> </ul> |
| <p>May 2022 – Aug 2022</p> <p>Strategic Command Bridges — Ain Sokhna Road Intersection Bridge No. 4</p>                     | <p>Jan 2022 – May 2022</p> <p>Sidi Abd El-Rahman Bridge, International Coastal Road</p> <p>The intersection of the international coastal road with Wadi El-Natron road</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>Jun 2021 – Dec 2021</p> <p>Recreational area in El Alamein City — second phase</p>                                       | <p>Feb 2021 – Jun 2021</p> <p>Jehan Sadat Corridor Development Project — First Phase, Third Sector</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| EGYPTIAN ARMED FORCES ENGINEERING AUTHORITY                                                                                                                                            |  |
| <p>Mar 2019 – Feb 2020</p> <p>Eastern Port Said Naval Base</p> <p>Dredging, ports, and marine works · Owner: Egyptian Navy · Consultants: Dar Al-Handasah and ACE Moharram-Bakhoun</p> |  |

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| TOOLS                                                                                                                                                   |                                                                                                                                                                                                                                                                   |
| <p><u>mbiX Xadd-in for Excel</u></p> <p>An Excel VSTO add-in that syncs Google Sheets data into a local SQLite database through a full ETL pipeline</p> | <p><u>ScrapeX for Chrome</u></p> <p>A Chrome extension and local engine that collects web data into a SQLite warehouse feeding the mbiX add-in</p>                                                                                                                |
| <p><u>Redundant relationship detection</u></p> <p>An algorithm that flags redundant logic links in Primavera P6 schedules</p>                           | <p><u>Spiral Reinforcement Detailing Sheet — Shape Code 77</u></p> <p>Computes helical reinforcement geometry and the number of spiral bars for a distribution length (BS 8666:2005), with the steps to produce shop drawings in AutoCAD Structural Detailing</p> |

# Education

Degrees and volunteering

| DEGREES                                                                                                                                                                                                                                                                                                                                                |
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| <div>Oct 2024 – Oct 2027 · in progress</div> <div><b>Master's degree, Data Science</b></div> <div>Cairo University</div>                                                                                                                                                                                                                               |
| <div>2013 – 2018</div> <div><b>Bachelor's degree, Civil Engineering</b></div> <div>Pyramids Higher Institute for Engineering and Technology (PHI), 6th of October</div> <div><ul style="list-style-type: none"><li>Graduation project in geotechnical engineering and foundations (A+)</li><li>Activities: Enactus PHI multimedia head</li></ul></div> |

| VOLUNTEERING                                                                                                                                                                                                                                                                                                                  |
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| <div>2016 – 2018</div> <div><b>Enactus PHI Multimedia Head</b></div> <div>Enactus · Economic empowerment</div> <div><ul style="list-style-type: none"><li>Study entrepreneurship and apply it in different projects</li><li>Volunteer as graphic designer, motion graphics artist, video editor, and marketer</li></ul></div> |

# Certifications

Licenses and certificates, newest first

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|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Feb 2025</div> <div><b>Certificate Professional Accreditation</b></div> <div>Saudi Council of Engineers</div>              | <div>Jan 2025</div> <div><b>ITIDA Gigs Program</b></div> <div>Information Technology Industry Development Agency (ITIDA) · Freelancing program</div> |
| <div>Oct 2024</div> <div><b>Google Data Analytics Professional Certificate</b></div> <div>Google</div>                          | <div>Sep 2024</div> <div><b>SQL</b></div> <div>HackerRank</div>                                                                                      |
| <div>Aug 2024</div> <div><b>Microsoft Power BI Data Analyst Professional Certificate</b></div> <div>Microsoft</div>             | <div>Aug 2024</div> <div><b>Tableau Business Intelligence Analyst Specialization</b></div> <div>Tableau</div>                                        |
| <div>Jul 2024</div> <div><b>Data Visualization with Tableau Specialization</b></div> <div>University of California, Davis</div> | <div>Jul 2024</div> <div><b>Data Analysis and Visualization Foundations Specialization</b></div> <div>IBM</div>                                      |
| <div>Jul 2024</div> <div><b>Delivering Quality Work with Agility</b></div> <div>IBM</div>                                       | <div>Jun 2023</div> <div><b>Navisworks Manage</b></div> <div>Autodesk</div>                                                                          |
|                                                                                                                                 |                                                                                                                                                      |

Jun 2023

**BIM Coordinator Diploma**

Autodesk

Mar 2022 · expires Mar 2027

**IPMA Level D® — Certified Project Manager Associate**

International Project Management Association (IPMA)

## Skills

Tendering, engineering, tools, software, and people

### Tendering and cost

BILL OF QUANTITIES (BOQ) QUANTITY SURVEYING CONSTRUCTION ESTIMATING COST CONTROL BID PREPARATION  
BID PRICING PRICING STRATEGY PRICING ANALYSIS TENDER PREPARATION TENDER SUBMISSIONS  
INVITATION TO TENDER PROJECT BIDDING FINANCIAL DATA ANALYSIS

### Engineering

TECHNICAL OFFICE TECHNICAL SPECS COMPUTER-AIDED DESIGN (CAD) BIM MODELLING CONSTRUCTION

### Tools and technologies

AUTOCAD REVIT NAVISWORKS ORACLE PRIMAVERA MICROSOFT PROJECT SAP ERP MICROSOFT EXCEL  
MICROSOFT POWERPOINT MICROSOFT WORD MICROSOFT POWER BI TABLEAU SQL

### Software development

C# / .NET VSTO EXCEL INTEROP SQLITE ETL PIPELINES GOOGLE SHEETS API PYTHON HTML / CSS / JS

### Interpersonal

TEAM LEADERSHIP SUPERVISORY SKILLS NEGOTIATION COMMUNICATION TEAMWORK PROBLEM SOLVING  
CRITICAL THINKING ANALYTICAL SKILLS ATTENTION TO DETAIL ETHICAL DECISION MAKING

## Languages

Spoken and written

Arabic

**Native or bilingual proficiency**

English

**Elementary proficiency**